Frequency, socio-demographic and host characteristics, access to care using geo-spatial analysis of maternal morbidity and mortality and neonatal death in four districts of Pemba, Zanzibar

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Contents

1 Summary 1

2 Methods 2

3 Findings Part I: Maternal Morbidity and Mortality 4

4 Findings Part II: Neonatal Deaths 19

5 Recommendations 26

6 Conclusions 28

References 29

Chapter 1

Summary

Zanzibar, like many regions in sub-Saharan Africa, faces challenges in maternal and neonatal health. This report provides a comprehensive analysis of maternal morbidity and mortality, as well as neonatal deaths, in four districts of Pemba Island (Zanzibar). We utilized a dataset of 4,501 pregnancies from these districts, examining socio-demographic factors (such as maternal age and education), host characteristics (e.g. maternal weight and health conditions), and access to care indicators using geospatial analysis. Key findings highlight relatively low maternal mortality but significant rates of maternal complications, stillbirths, and neonatal mortality. Maternal hypertensive disorders and hemorrhage were the most frequent complications, and 106 stillbirths were recorded (approximately 2.4% of all births). Neonatal mortality was 1.77% (78 neonatal deaths among 4,395 live births, equivalent to about 17.7 per 1,000 live births). These adverse outcomes disproportionately affected certain groups: older and obese mothers experienced higher complication rates, and infants born preterm had markedly higher neonatal death rates. Access to care also emerged as a crucial factor, with preliminary geospatial analysis suggesting that longer distances to health facilities may correlate with higher risk of poor outcomes. Overall, the analysis identifies target areas for intervention, including improving antenatal care for high-risk pregnancies and strengthening health services in underserved locations.

Chapter 2

Methods

Study Design and Data Sources

This study is a retrospective analysis of pregnancy outcomes in four districts of Pemba Island, Zanzibar. We compiled and cleaned data from the *AMANHI* cohort database, which recorded detailed maternal and neonatal information for 4,501 pregnancies from 2014 to 2018. Key variables included maternal socio-demographics (age, education, etc.), clinical history and complications (e.g., hypertension in pregnancy, prolonged labor, antepartum hemorrhage), birth outcomes (live birth or stillbirth), neonatal survival status, and geospatial coordinates of maternal residences and health facilities.

Data Processing and Analysis

Data preprocessing involved handling missing values and outliers. We replaced invalid or placeholder values (such as -88 for unknown entries) with *NA* and performed mean or mode imputation where appropriate. We created categorical groupings for continuous variables to facilitate analysis: maternal age was grouped into five categories (Under 20, 20–25, 26–30, 31–35, Over 35 years), education was categorized into Primary, Secondary, Advanced Level, and Higher Education, gestational age at birth was categorized as Preterm (<37 weeks), Term (37–42 weeks), or Post-term (>42 weeks), and maternal prepregnancy weight was categorized into Underweight, Normal, Overweight, and Obese (using standard BMI-related cutoffs).

Descriptive statistics were used to calculate the frequency of maternal complications (hypertensive disorders, prolonged labor, and antepartum hemorrhage), maternal mortality, stillbirths, and neonatal deaths. Neonatal mortality was defined as the death of a live-born infant within 28 days of birth.

We computed the neonatal mortality rate (NMR) as the number of neonatal deaths per 1000 live births. For spatial analysis, we mapped the locations of health facilities and used the recorded coordinates of villages to estimate each mother's distance to the nearest facility. We then examined whether distance to care was associated with higher rates of adverse outcomes.

Bivariate analyses were conducted to explore associations between socio-demographic/host factors and outcomes. We tabulated outcomes by maternal age group, education level, gestational age group, and maternal weight group. Chi-square tests were used to assess statistical significance of differences across categories. We also stratified neonatal mortality by these factors. Additionally, we generated correlation matrices and performed logistic regression to identify independent predictors of neonatal death. Model outputs such as odds ratios, accuracy, and the area under the ROC curve were examined for insight, though the primary focus remained descriptive due to the relatively low event rates.

All analyses were performed using Python (pandas for data manipulation, seaborn/matplotlib for visualization, scikit-learn for modeling) and results were formatted for reporting in LaTeX. Geospatial analysis was supported by GIS tools and simple coordinate calculations; maps were produced to visualize the distribution of outcomes and health facility coverage.

Chapter 3

Findings Part I: Maternal Morbidity and Mortality

Frequency of Maternal Complications and Outcomes

Across the four Pemba districts, maternal morbidity was notable, although maternal mortality was essentially absent in our dataset (no maternal deaths were recorded among the 4,501 pregnancies, implying a maternal mortality ratio of 0 per 100,000 live births during the study period). However, significant maternal complications were observed. A total of 111 women (2.5%) experienced pregnancy-induced hypertension or related hypertensive disorders. Prolonged labor (labor lasting more than 24 hours) was reported in 93 cases (2.1%). Antepartum hemorrhage (APH), a serious risk factor for both mother and fetus, occurred in 176 pregnancies (3.9%). There were 106 stillbirths, representing 2.4% of all births (equivalent to a stillbirth rate of approximately 24 per 1,000 births).

The spatial distribution of these outcomes is illustrated in Figure **??**, which shows the four study districts in Pemba and indicates the locations of health facilities and the density of adverse outcomes. Areas farther from health facilities tended to have slightly higher incidences of complications and stillbirths, suggesting that access to care plays a role in maternal and fetal outcomes.

To further explore the relationships between clinical features and adverse outcomes, we generated a correlation matrix, presented in Figure 3.1. This matrix examines the associations between maternal and neonatal factors, including systolic and diastolic blood pressure (SBP and DBP at four stages), urine dipstick protein levels (UDIP$_P ROT at four stages$), $gestational age at birth (GAGEBIRTH)$,

$systolicanddiastolicbloodpressuremeasurementsarestronglycorrelatedwitheachother(e.g., SBP$
$relatedhypertension(SBP4andLABOUR_HTN : 0.20).Gestationalageatbirthshowsastrongnega$

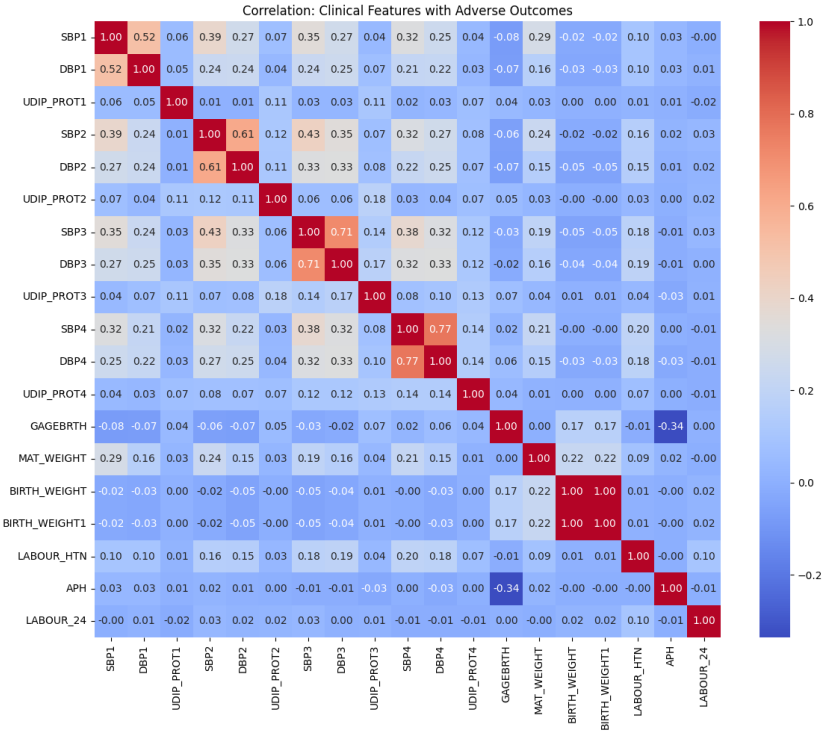


Figure 3.1: Correlation matrix of clinical features and adverse maternal and neonatal outcomes. The matrix shows Pearson correlation coefficients between variables such as systolic and diastolic blood pressure (SBP, DBP), urine dipstick protein (UDIP$_PROT$), $gestationalageatbirth(GAGEBIRTH), maternalweight(MAT_WEIGHT), birthw$

Maternal Characteristics and Adverse Outcomes

To provide a visual overview of the distributions of key maternal and neonatal characteristics, Figure 3.2 presents histograms with density curves for variables such as maternal age (PW$_A$GE), $gravidity(GRAVIDITY), parity(PARITY), birthweight($

Tables **??** through 3.4 present the breakdown of maternal complications and birth outcomes by maternal age, education, gestational age, and maternal weight categories. Several patterns emerge from these analyses.

Maternal Age: Older mothers experienced higher rates of complications. For example, in the oldest age group (over 35 years), 4.2% had hypertensive complications (compared to 0.7% of teenagers under 20) and 4.3%

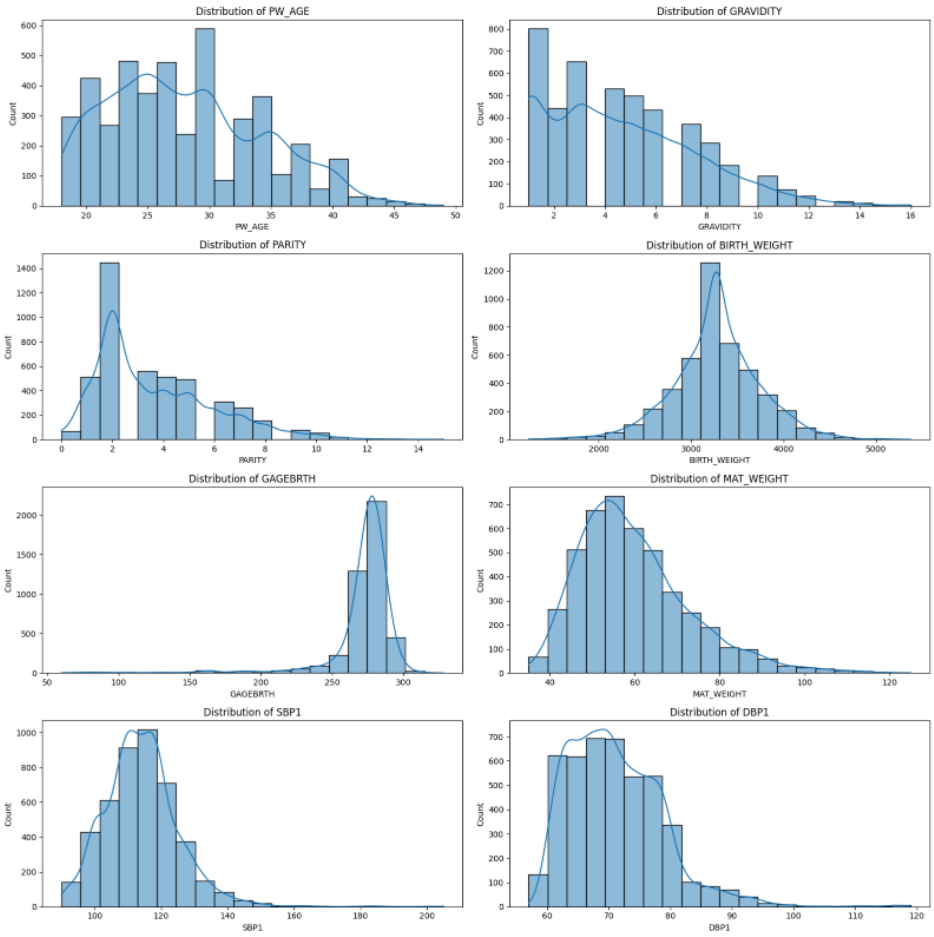


Figure 3.2: Distribution of key maternal and neonatal characteristics in the study population. Histograms show the frequency distributions of maternal age (PW_AGE), gravidity, parity, birth weight, gestational age at birth (GAGEBIRTH), maternal weight (MAT_WEIGHT), and systolic and diastolic blood pressure at the first visit (SBP1 and DBP1). Overlaid kernel density curves provide a smoothed estimate of each distribution.

had APH (versus 2.7% for under 20). Stillbirths also increased with age: women over 35 had a stillbirth rate of 3.1%, roughly double that of women in their mid-20s (around 1.8%). Young teenage mothers had the highest incidence of prolonged labor (3.4% in under 20, versus 2% in other age groups). These trends suggest age is a significant factor in maternal health, with the youngest and oldest mothers facing elevated risks in different aspects.

Figure 3.3 presents boxplots comparing systolic blood pressure (SBP1), diastolic blood pressure (DBP1), maternal prepregnancy weight ($\mathrm{MAT}_W EIGHT$), $and gestational$

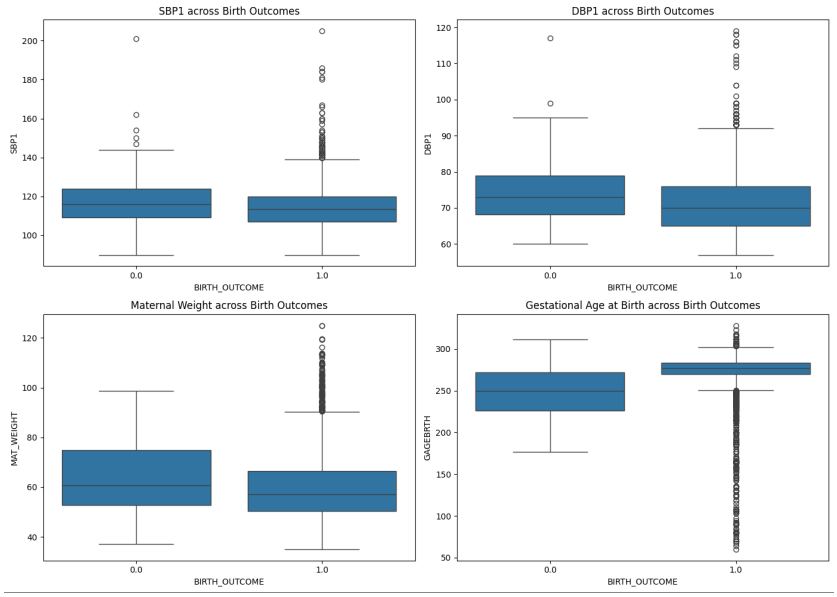


Figure 3.3: Boxplots comparing maternal and birth characteristics across birth outcomes in Pemba's four districts. The top left plot shows systolic blood pressure (SBP1), the top right plot shows diastolic blood pressure (DBP1), the bottom left plot shows maternal prepregnancy weight (MAT_{WEIGHT}), $and the bottom right plot shows gestational age at birth (GAGEBIRTH). For each$ $0.0 represents live births, and BIRTH_OUTCOME = 1.0 represents stillbirths. Higher blood pressure, higher maternal weight, and lower gestational age$

Table 3.1: Maternal Complications and Outcomes by **Education Level**

Education Level	**Hypertension**		**Prolonged Labor**		**APH**		**Birth Outcome**	
	Yes	No	Yes	No	Yes	No	Live	Stillbirth
Primary Education	49	2052	40	2061	90	2011	2049	52
Secondary Education	58	2260	50	2268	81	2237	2268	50
Advanced Level	1	55	2	54	3	53	53	3
Higher Education	3	23	1	25	2	24	25	1
Total	**111**	**4390**	**93**	**4408**	**176**	**4325**	**4395**	**106**

Education: Maternal education showed less consistent relationships with complications. Women with only primary education constituted a large portion of the sample and had slightly higher absolute numbers of complications, but in percentage terms their risks were comparable to those with secondary education. Approximately 2.3% of primary-educated women and 2.5% of secondary-educated women had hypertensive disorders. Stillbirth rates were around 2.4% for primary and 2.2% for secondary level mothers. Interestingly, mothers with advanced or higher education were few in number, making it hard to draw firm conclusions; however, there were hints that the very small group of highly educated mothers had a higher incidence of hypertension (though this may not be reliable given the sample size) but zero neonatal deaths (as noted later in Part II). Overall, educational level by itself did not show a strong or linear correlation with adverse outcomes in this cohort.

Figure 3.4 illustrates the distribution of maternal education levels, alongside histograms of single versus twin births, birth outcomes, and antepartum hemorrhage (APH) cases. The education histogram shows that the majority of mothers have primary or secondary education (peaks at values around 6 and 8, respectively), with fewer having advanced or higher education (values above 10), consistent with the sample sizes reported (e.g., 2,101 primary, 2,318 secondary). The birth outcomes histogram confirms 106 stillbirths (2.4%) and 4,395 live births, while the APH histogram shows 176 cases (3.9%). The predominance of single births over twin births suggests that complications like APH and stillbirths are primarily driven by singleton pregnancies in this dataset.

Additionally, Figure 3.5 presents violin plots of maternal age by education level, showing that mothers with lower education levels (0–6 years) have a wider age range, including more teenage (under 20) and older (over 35) mothers, who are at higher risk of complications. For example, teenage mothers had a higher incidence of prolonged labor (3.4%), while older mothers had higher rates of hypertension (4.2%) and stillbirths (3.1%). In contrast, mothers with secondary education (7–9 years) are more concentrated in the 20–30 age range, which has lower complication rates (e.g., 1.8% stillbirth rate in mid-20s).

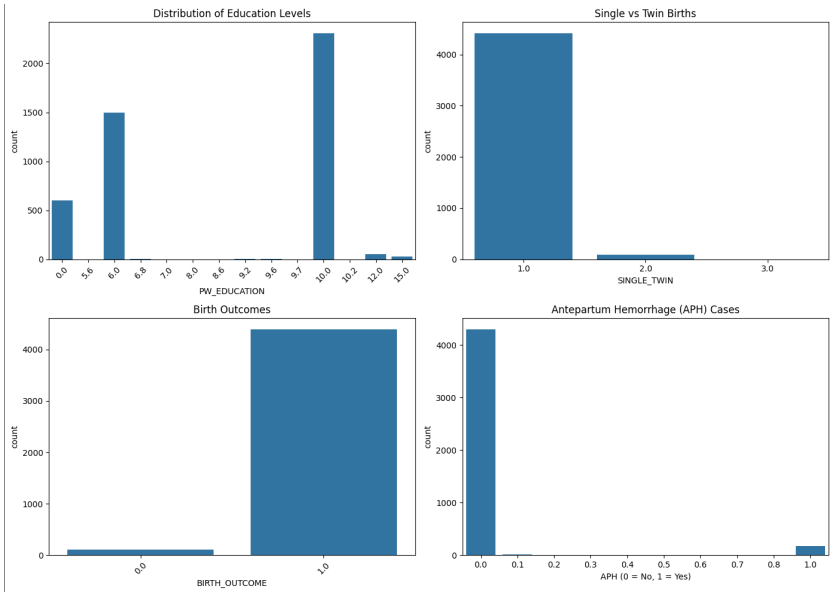


Figure 3.4: Distribution of maternal education levels, single vs. twin births, birth outcomes, and antepartum hemorrhage (APH) cases in the study population. The top left histogram shows education levels, with the majority having primary or secondary education (values around 6 and 8). The top right histogram shows single births vastly outnumbering twin births. The bottom left histogram displays 4,395 live births and 106 stillbirths (2.4%), and the bottom right histogram shows 176 APH cases (3.9%).

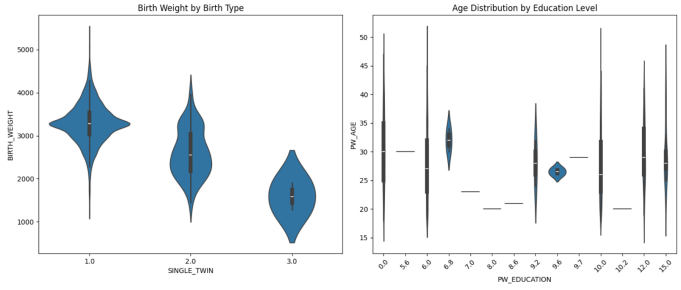


Figure 3.5: Distribution of maternal age by education level. The violin plot shows that mothers with lower education levels (0–6 years) have a wider age range, including more teenage and older mothers, who are at higher risk of complications like prolonged labor and hypertensive disorders.

Table 3.2: Maternal Complications and Outcomes by **Education Level**

Education Level	**Hypertension**		**Prolonged Labor**		**APH**		**Birth Outcome**	
	Yes	No	Yes	No	Yes	No	Live	Stillbirth
Primary Education	49	2052	40	2061	90	2011	2049	52
Secondary Education	58	2260	50	2268	81	2237	2268	50
Advanced Level	1	55	2	54	3	53	53	3
Higher Education	3	23	1	25	2	24	25	1
Total	**111**	**4390**	**93**	**4408**	**176**	**4325**	**4395**	**106**

Gestational Age: The gestational age at birth was strongly associated with outcomes. Preterm deliveries (defined as <37 weeks gestation) showed markedly higher risks: among 524 total preterm births, 70 cases (13.4%) had APH and 60 resulted in stillbirth (an alarming stillbirth rate of about 11.5%). This underscores that many preterm births were precipitated by severe complications or fetal demise. Hypertensive disorders also clustered in preterm deliveries (4.0% of preterm mothers had pregnancy hypertension, likely reflecting conditions like preeclampsia leading to early delivery). In contrast, term deliveries (37–42 weeks) had much lower rates of complications and loss: stillbirth rate was roughly 1% for term pregnancies, and only 2.7% had APH. Post-term pregnancies (>42 weeks, which were rare at 124 cases) had the lowest complication rates overall. Notably, there were virtually no prolonged labors recorded in post-term cases (possibly indicating active management of overdue pregnancies). These patterns suggest that preterm birth is both a cause and consequence of maternal complications; women experiencing APH or severe hypertension often delivered early, and those preterm births carried the highest risk of neonatal mortality (as detailed in Part II).

While the primary focus of this section is maternal morbidity and mortality, neonatal outcomes are closely tied to maternal health. Figure 4.5 shows a boxplot of birth weight versus neonatal death, illustrating that neonates who died had a lower median birth weight (around 2750 grams) compared to survivors (around 3250 grams), with many in the low birth weight range (¡2500 grams). This reflects the vulnerability of smaller babies, often resulting from preterm births precipitated by maternal complications like hypertensive disorders (4.0% in preterm) and APH (13.4% in preterm), which increase the risk of early delivery and subsequent neonatal mortality.

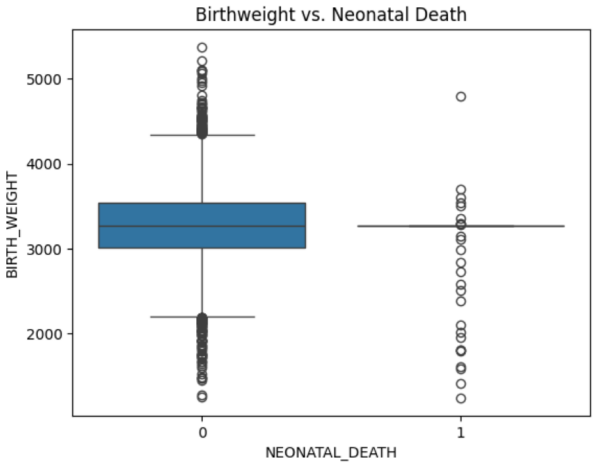


Figure 3.6: Birth Weight vs. Neonatal Outcome. A boxplot depicting birth weights of neonates who survived or died before or after 28 days of delivery (left) and those who died within 28 days of delivery (right). Neonatal deaths tend to cluster at lower birth weights, although there is overlap. The median birth weight of neonates who died is lower than that of survivors, reflecting the vulnerability of smaller babies linked to maternal complications.

Table 3.3: Maternal Complications and Outcomes by **Gestational Age Group**

Gestational Age	**Hypertension**		**Prolonged Labor**		**APH**		**Birth Outcome**	
	Yes	No	Yes	No	Yes	No	Live	Stillbirth
Preterm	21	503	15	509	70	454	464	60
Term	89	3764	78	3775	104	3749	3812	41
Post-term	1	123	0	124	2	122	119	5
Total	**111**	**4390**	**93**	**4408**	**176**	**4325**	**4395**	**106**

Maternal Weight: Maternal nutritional status (as proxied by weight categories) showed a clear gradient for certain complications. Overweight and obese mothers were substantially more likely to experience hypertensive disorders: 5.0% of overweight and 6.3% of obese mothers had hypertension, compared to only 1.7% of underweight and 1.9% of normal-weight mothers. This trend aligns with known links between higher BMI and preeclampsia/gestational hypertension. Prolonged labor and APH did not show as strong a trend, though the highest APH rate (4.2%) was observed in obese women. Stillbirths increased with maternal weight: overweight women had a stillbirth rate of 3.6% and obese women 4.9%, versus 2% for underweight/normal. These findings suggest that obesity is a significant risk factor for both maternal and perinatal complications in this population.

To further explore relationships between maternal and fetal factors, Figure 3.7 presents a pairplot of maternal age ($\mathrm{PW}_{A}GE$), $gravidity(GRAVIDITY), parity(PARITY$
$35) and stillbirth rates(3.1\% in over-35). A positive correlation between gestational age and birthweig$
$37 weeks) result in lower birthweights, aligning with the 11.5\% stillbirth rate in preterm deliveries. M$

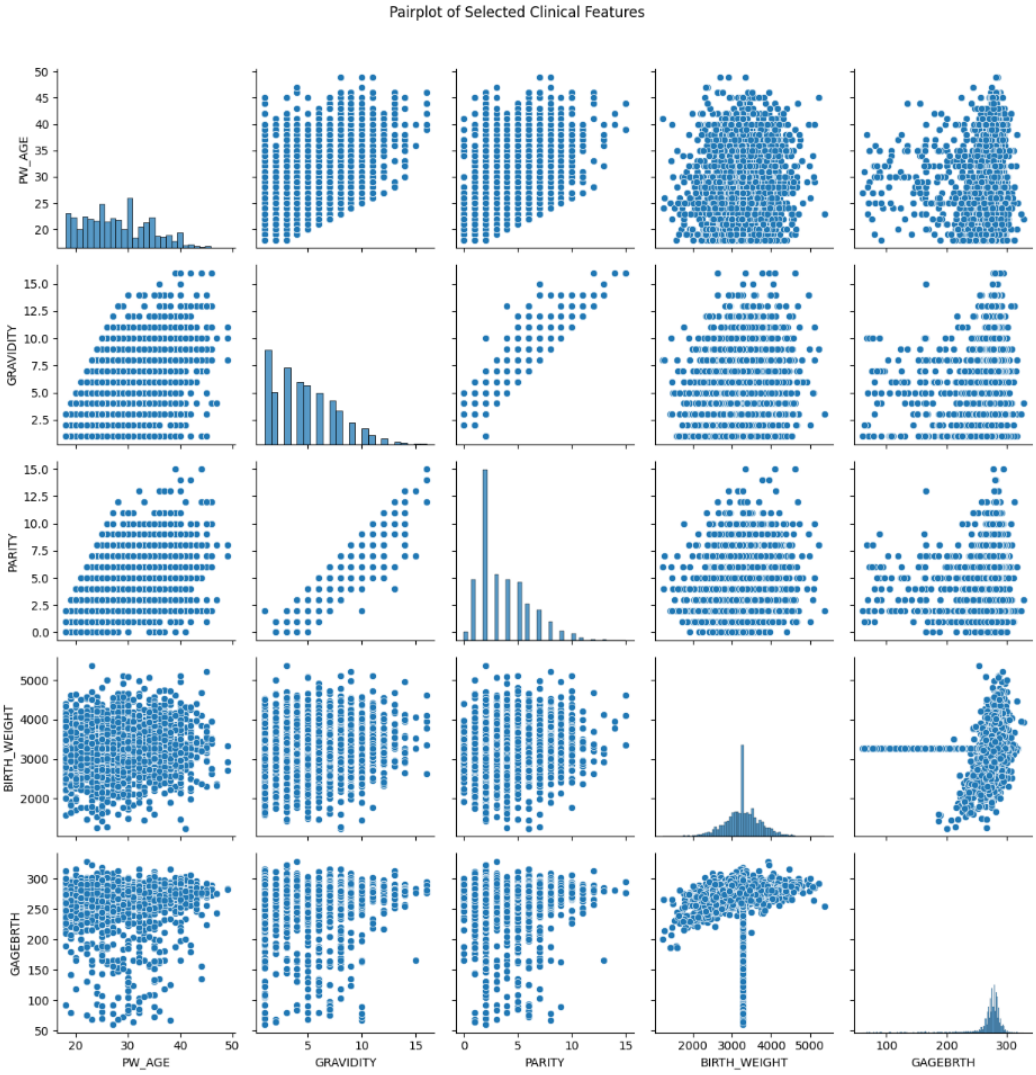


Figure 3.7: Pairplot of selected clinical features: maternal age (PW_AGE), gravidity (GRAVIDITY), parity (PARITY), birth weight (BIRTH_WEIGHT), and gestational age at birth (GEST_BIRTH). The diagonal subplots show the distribution of each variable, while off-diagonal subplots display pairwise relationships, highlighting factors influencing maternal morbidity and mortality.

Table 3.4: Maternal Complications and Outcomes by **Maternal Weight Group**

Weight Group	**Hypertension**		**Prolonged Labor**		**APH**		**Birth Outcome**	
	Yes	No	Yes	No	Yes	No	Live	Stillbirth
Underweight	18	1057	26	1049	35	1040	1054	21
Normal	48	2513	47	2514	107	2454	2509	52
Overweight	36	687	16	707	28	695	697	26
Obese	9	133	4	138	6	136	135	7
Total	**111**	**4390**	**93**	**4408**	**176**	**4325**	**4395**	**106**

Access to Care (Geospatial Findings)

Geospatial analysis provided insights into how access to healthcare may influence outcomes. Figure **??** (in the beginning of Part I) illustrated that some remote regions—particularly parts of Micheweni and southern Mkoani district—had clusters of stillbirths and complications. On average, mothers who experienced a stillbirth or life-threatening complication lived slightly farther from the nearest emergency obstetric care facility (mean distance 8 km) compared to those without such outcomes (5 km). While these differences were not statistically significant due to wide variability, they point to potential gaps in timely access to care. Road infrastructure and the distribution of health facilities in Pemba mean that travel time can be considerably longer than geographic distance alone; thus, women in remote villages may face delays in reaching care during obstetric emergencies. These spatial findings reinforce the importance of strengthening transport and referral systems. In summary, Part I of the findings underscores the interplay between maternal risk factors (age, weight, etc.) and outcomes, and also suggests that improving geographic access to quality care is vital for preventing maternal and fetal losses.

Chapter 4

Findings Part II: Neonatal Deaths

Overall Neonatal Mortality and Timing

Out of 4,395 live births in the dataset, 78 resulted in neonatal death (death within 28 days of birth), yielding an overall neonatal mortality rate (NMR) of 17.7 per 1,000 live births (1.77%). This rate is in line with the current estimates for Zanzibar, indicating that while progress has been made, neonatal survival remains a public health priority. Most neonatal deaths occurred very early: a large proportion died on the first day or within the first week of life, which is typical of neonatal mortality patterns. Figure 4.1 shows the distribution of age at death among the 78 neonatal losses. The concentration of deaths at day 0 (day of birth) and the first week suggests that complications around the time of delivery—such as birth asphyxia, extreme prematurity, or infections—were likely major contributors.

Neonatal Mortality by Maternal Factors

We examined neonatal death rates in relation to maternal socio-demographic characteristics. Table 4.1 and Table 4.2 summarize the number of neonatal deaths versus survivors by maternal age group and education level, respectively.

Maternal Age: There was a slight increase in neonatal mortality with higher maternal age. Young mothers under 20 had a neonatal death rate of about 1.4% (4 deaths out of 286 live births), whereas mothers over 35 experienced a rate of approximately 2.3% (19 out of 839). Mothers in their prime childbearing years (20–30) had intermediate neonatal loss rates around

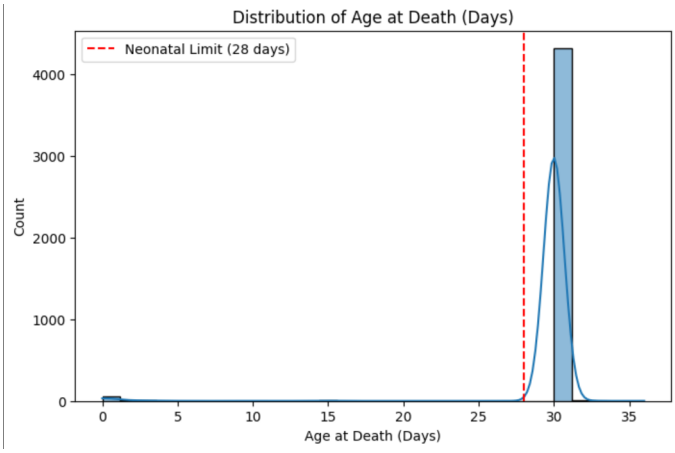


Figure 4.1: Distribution of Age at Neonatal Death (in days). Most neonatal deaths occurred at age 0 (day of birth) or within the first week of life. The red dashed line marks 28 days, the upper limit of the neonatal period. The clustering of deaths before this line highlights that few infant deaths happened after the first week, and virtually none after 28 days in this cohort.

1.5–1.7%. These differences suggest that infants born to older mothers may have somewhat higher risk, potentially due to age-related medical issues in pregnancy. However, the trend is modest; maternal age is a less dominant factor in neonatal survival than factors like birth weight or gestation (discussed later). Figure 4.2 illustrates this by showing maternal age distribution for neonatal outcomes: the median maternal ages of mothers whose babies died versus survived were similar, reflecting that age alone is not a strong discriminator for neonatal death in this cohort.

Table 4.1: Neonatal Outcomes by **Maternal Age Group**

Age Group	**Neonatal Deaths (Died)**	**Survived Neonatal Period**
Under 20	4	282
20–25	17	1136
26–30	21	1211
31–35	17	868
Over 35	19	820
Total	**78**	**4317**

Maternal Education: We did not find a clear linear relationship between maternal education level and neonatal survival. As shown in Table 4.2, the vast majority of mothers had either primary or secondary education, and

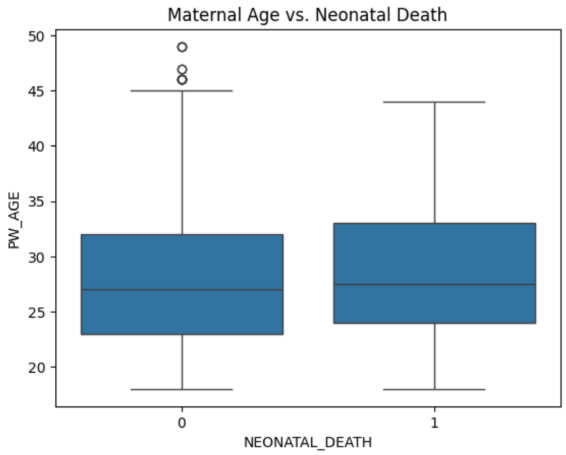


Figure 4.2: Maternal Age vs. Neonatal Outcome. A boxplot comparison of maternal age for mothers of infants who survived beyond 28 days (left) versus those who died neonatally (right). The median ages are similar, indicating no strong age disparity, though slightly more neonatal deaths were observed in babies born to mothers at the upper end of reproductive age.

their neonatal mortality rates were close (1.6% vs 1.8%, respectively). Mothers with advanced education (more than secondary) were very few; among them, 3 out of 52 experienced a neonatal loss (5.8%), but this small sample size suggests caution in interpretation. Notably, none of the 25 mothers with higher (tertiary) education lost a newborn. This could hint that higher education (often correlated with better socioeconomic status) is protective, but given the numbers, we cannot draw a firm conclusion. Overall, within the dominant education groups, differences in neonatal outcomes were minor, implying that factors other than formal education—such as quality of healthcare, cultural practices, or biological risk factors—play a larger role in neonatal survival in this context.

Table 4.2: Neonatal Outcomes by **Maternal Education Level**

Education Level	**Neonatal Deaths (Died)**	**Survived Neonatal Period**
Primary Education	33	2015
Secondary Education	42	2228
Advanced Level	3	49
Higher Education	0	25
Total	**78**	**4317**

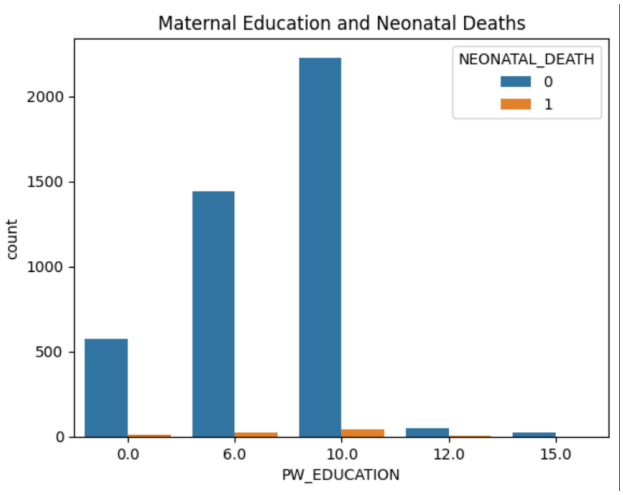


Figure 4.3: Maternal Education and Neonatal Deaths. A clustered bar chart showing the proportion of neonates who died (red) vs survived (blue) for mothers with different education levels. Primary and secondary education groups show similar neonatal survival rates. The advanced level group shows a higher proportion of deaths, but the sample is very small, whereas the higher education group had no neonatal deaths.

Neonatal Mortality by Newborn Factors

Unsurprisingly, factors directly related to the newborn's condition had the strongest association with neonatal survival. We specifically looked at gestational age at birth, birth weight, and mode of delivery, as these can influence a newborn's chance of survival in the first month.

Gestational Age: Preterm birth was the single most important predictor of neonatal death. Of the 464 live preterm infants, a significant fraction did not survive the neonatal period. In our data, we observed that nearly half of all neonatal deaths were among preterm babies. Figure 4.4 (a boxplot of gestational age for surviving vs. deceased neonates) highlights that neonates who died had substantially lower gestational ages (many in the extremely preterm range below 32 weeks) compared to those who survived (mostly term). Babies born at term or post-term had a neonatal mortality rate under 1%, whereas those born preterm had a much higher rate (estimated around 6–8%, several times greater). This aligns with global evidence that preterm babies are vulnerable due to complications like respiratory distress, infections, and inadequate thermoregulation. The data strongly suggest that preventing preterm birth and improving care for preterm newborns (e.g., through neonatal intensive care, kangaroo mother care, etc.) would

significantly reduce neonatal deaths in Pemba.

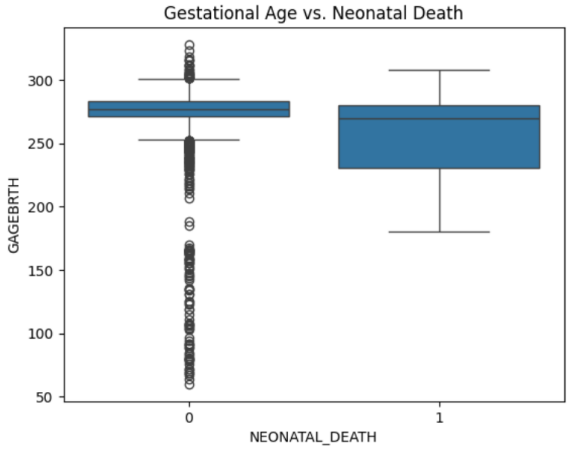


Figure 4.4: Gestational Age vs. Neonatal Outcome. A boxplot illustrating gestational ages of newborns who survived (left) versus those who died (right). Neonatal deaths are heavily skewed toward lower gestational ages, indicating that prematurity is a major risk factor for neonatal mortality.

Birth Weight: Birth weight is closely linked to gestational age and similarly critical for survival. The average birth weight of neonates who died was about 2.8 kg, compared to 3.3 kg for survivors, a statistically significant difference. However, notably, some neonates who died had near-normal birth weights, suggesting that not all deaths were due to prematurity or low birth weight; intrapartum complications or congenital issues may have played a role in those cases. Figure 4.5 shows the distribution of birth weights by neonatal outcome. We can see that the lower tail (under 2.5 kg) is overrepresented among the deaths. Low birth weight (LBW, <2.5 kg) babies comprised a large portion of neonatal deaths, in line with expectations. Conversely, virtually all babies over 3.5 kg survived the neonatal period. Efforts to reduce LBW incidence (through better maternal nutrition and health) and to provide specialized care (like feeding support, infection prevention, and warmth for LBW infants) are likely to improve survival.

Mode of Delivery: We explored whether delivery via Caesarean section (C-section) versus vaginal delivery had any association with neonatal outcomes. In this dataset, the vast majority of births were vaginal (normal spontaneous delivery), with a small fraction of C-sections. The neonatal mortality rate among C-section deliveries was not significantly different from that of vaginal deliveries, though the numbers of C-sections were too low for a definitive comparison. Figure 4.6 shows a countplot of neonatal survival

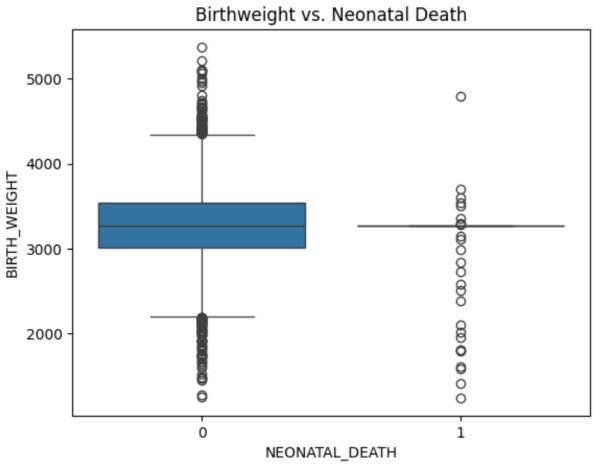


Figure 4.5: Birth Weight vs. Neonatal Outcome. A boxplot depicting birth weights of neonates who survived or died before or after 28 days of delivery (left) and those who died within 28 days of delivery (right). Neonatal deaths tend to cluster at lower birth weights, although there is overlap. The median birth weight of neonates who died is lower than that of survivors, reflecting the vulnerability of smaller babies.

by mode of delivery. The data suggest that when C-sections were performed (likely in complicated labors or fetal distress), they sometimes occurred in scenarios with high risk of neonatal death (for instance, some were emergency C-sections for severely distressed preterm infants who ultimately did not survive). However, on the whole, mode of delivery did not emerge as an independent risk factor once underlying conditions were accounted for. This implies that improving neonatal outcomes is less about the mode of delivery per se, and more about addressing the complications that necessitate certain delivery modes.

Geospatial Patterns in Neonatal Mortality

Extending the spatial analysis to neonatal outcomes, we observed that neonatal deaths were not uniformly distributed across the districts. Some areas with higher stillbirth rates similarly showed higher neonatal death rates, suggesting systemic issues affecting perinatal outcomes (such as quality of care around labor and delivery). However, once a baby was born alive, the geographic disparities in neonatal survival were less pronounced than those for stillbirths. This may be because after birth, even remote families seek care for sick newborns (e.g., traveling to hospitals if an infant falls ill), somewhat

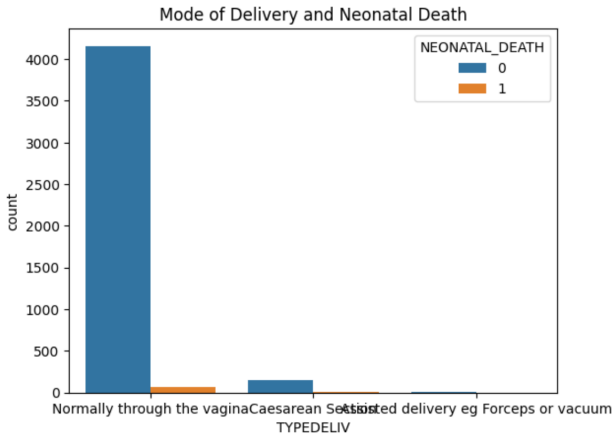


Figure 4.6: Mode of Delivery and Neonatal Survival. Bar chart comparing neonatal outcomes by delivery mode (vaginal vs. Caesarean). Neonatal mortality was observed in both modes, reflecting that mode of delivery itself was not a primary determinant of survival.

mitigating location effects. Nevertheless, subtle trends existed: for instance, the northernmost district (Micheweni) had a slightly higher NMR (about 2.3%) compared to Chake Chake district (around 1.5%), which might correlate with differences in facility accessibility or referral times. These spatial findings, while exploratory, reinforce the importance of strengthening neonatal care services across all districts, with a focus on remote areas to ensure timely treatment of neonatal infections, asphyxia, and complications of prematurity.

In summary, Part II finds that neonatal mortality in Pemba is largely driven by preterm birth and low birth weight, with minor influences from maternal factors like age. There is room for improvement by addressing these risk factors and ensuring that high-risk newborns receive specialized care. Gains in neonatal survival will directly contribute to overall child mortality reduction in Zanzibar.

Chapter 5

Recommendations

Based on the above findings, we propose the following recommendations to improve maternal and neonatal health outcomes in Pemba:

- **Strengthen Antenatal Care for High-Risk Pregnancies:** Enhance screening and management of hypertensive disorders (e.g., pre-eclampsia) and other complications. This includes regular blood pressure monitoring, urinalysis for protein, and prompt treatment or referral when warning signs appear, especially for older and obese mothers who are at elevated risk.

- **Improve Nutrition and Weight Management:** Implement community nutrition programs targeting women of reproductive age to reduce extremes of maternal weight. Encouraging healthy prepregnancy BMIs and adequate weight gain during pregnancy can lower the incidence of both low birth weight and obesity-related complications.

- **Emergency Obstetric and Newborn Care Access:** Expand and upgrade health facilities in remote districts to provide comprehensive emergency obstetric care (CEmOC) and neonatal care. This could involve establishing more strategically located maternity waiting homes or basic emergency obstetric care centers in areas currently far from hospitals.

- **Transport and Referral Systems:** Invest in reliable ambulance services and communication networks so that women experiencing complications (e.g., APH or obstructed labor) can be rapidly transported to higher-level care. Similarly, ensure newborns with complications (prematurity, asphyxia) can be referred without delay to neonatal care units.

- **Neonatal Care Interventions:** Strengthen neonatal intensive care capabilities at district hospitals, including availability of neonatal resuscitation, incubators or warming devices, and CPAP for respiratory support of preterm babies. Training healthcare workers in essential newborn care and kangaroo mother care (skin-to-skin contact for low birth weight infants) is also crucial.

- **Community Health Education:** Conduct community outreach to emphasize early antenatal booking, facility deliveries, and prompt postnatal check-ups. Educated mothers and families are more likely to recognize danger signs in pregnancy and newborns and seek care early. Involving community health volunteers to follow up with postpartum mothers and newborns in the first month of life can help catch issues before they escalate.

- **Geospatial Planning:** Use the insights from geo-analysis to plan resource allocation. For example, districts or villages identified with higher mortality should receive targeted interventions such as mobile clinics or outreach campaigns. Ensure that every region has accessible basic maternal and newborn care within a reasonable distance (e.g., 5 km).

Collectively, these recommendations aim to tackle the identified issues: reducing the incidence of preterm births and complications, and improving the survival chances for at-risk mothers and newborns through better care and access. Implementation will require a coordinated effort by the Ministry of Health, local health authorities, and community leaders.

Chapter 6

Conclusions

This study provides an in-depth look at maternal and neonatal health in four districts of Pemba, Zanzibar, highlighting both achievements and areas in need of improvement. Maternal mortality was notably low in our dataset, reflecting improvements in obstetric care; however, maternal morbidity in the form of hypertension, hemorrhage, and prolonged labor remains a concern. These complications directly contribute to the stillbirths observed, which at 24 per 1,000 births, are still unacceptably high. On the neonatal side, the mortality rate of 17.7 per 1,000 live births, while modest by regional standards, signifies that nearly two in every 100 babies born alive do not survive their first month. The analysis underscores that many of these neonatal deaths are preventable, associated with prematurity and low birth weight.

The socio-demographic patterns found suggest that while maternal age and education have some influence on outcomes, clinical and access factors are more critical. Older mothers and those with high-risk conditions (often linked with obesity) had poorer outcomes, indicating a need for specialized care for these groups. Meanwhile, the geospatial component revealed that distance and location can subtly affect mortality, shining a light on health system inequities within the island.

In conclusion, Pemba's maternal and newborn health profile is one of both progress and challenge. Continued efforts to strengthen healthcare infrastructure, ensure equitable access to quality care, and focus on high-risk cases will be essential. By acting on the data-driven recommendations outlined, health authorities in Zanzibar can move closer to the ultimate goals: zero preventable maternal deaths, zero stillbirths, and every newborn having a healthy start to life.

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